

Automated Software Safety Scenario Matrix

(a) Explicitly asserted fail-safe behaviors

RC takeover	✓	—	✓	—	✓	✓
Emergency stop	✓	✓	✓	✓	—	—
Target lost	✓	✓	✓	✓	—	—
Required input stale	✓	✓	✓	✓	—	—
Base status fault/stale	✓	✓	✓	✓	—	—
Stale nonzero command	✓	—	—	—	—	—
Obstacle in stopping sweep	✓	—	—	—	—	—
	hazard / fault detected	authorization revoked	disarm requested	logical target reset	zero velocity command	no automatic resume

(b) Evidence layer and test coverage

RC takeover	ROS 2 launch · 2 tests
Emergency stop	ROS 2 launch · 1 test
Target lost	ROS 2 launch · 1 test
Required input stale	ROS 2 launch · 1 test
Base status fault/stale	ROS 2 launch · 2 tests
Stale nonzero command	C++ unit · 2 tests
Obstacle in stopping sweep	C++ unit · 1 test

Safety: 18/18 passed
Decision: 36/36 passed

SYMBOL

✓ explicitly asserted · — outside cited test scope

CAMPAIGN

54 individual tests · 0 failures

EXECUTION

ROS 2 launch + C++ unit + static contract tests

LIMIT

software evidence; not functional-safety certification